

Calculus with Parametric and Polar Curves



Parametric derivatives and arc length

- 1 For the curve $x = t^2$, $y = t^3$, find $\frac{dy}{dx}$ in terms of t .
 - 2 Find the arc length of the curve $x = 3t^2$, $y = 2t^3$ for $0 \leq t \leq 2$.
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Polar area

- 3 Find the area enclosed by one loop of the rose $r = 3\sin 2\theta$ (for $0 \leq \theta \leq \pi/2$).
- 4 Find the area enclosed by the cardioid $r = 2 + 2\cos \theta$.